Heart Attack Risk Prediction Using Machine Learning

Hackathon Submission

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Problem Statement

Cardiovascular diseases, particularly heart attacks, are one of the leading causes of death worldwide. Early identification of high-risk individuals can significantly reduce mortality rates through preventive care and timely medical intervention. However, manual risk assessment can sometimes be time-consuming and dependent on subjective clinical judgment.

The objective of this project is to develop a machine learning-based predictive model that can analyze patient health parameters and classify individuals based on their likelihood of experiencing a heart attack. By leveraging historical medical data and supervised learning algorithms, the model aims to assist healthcare professionals in early detection and risk stratification.

This solution focuses on building an accurate, interpretable, and practical system that can support data-driven decision-making in healthcare.

Heart Attack Risk Prediction Using Machine Learning

1. Introduction

Heart attacks are a major global health concern and contribute significantly to mortality rates worldwide. Early prediction and identification of high-risk individuals can help reduce complications through timely intervention and preventive care. Machine learning techniques provide an opportunity to analyze medical data and uncover patterns that may not be immediately visible through traditional analysis.

This project aims to build a predictive model that classifies patients based on their risk of experiencing a heart attack using clinical and demographic features.

2. Dataset Description

The dataset used in this project contains patient health records including demographic information and clinical measurements. Key features include:

- Age
- Sex
- Chest pain type
- Resting blood pressure
- Cholesterol level
- Maximum heart rate achieved
- Fasting blood sugar
- Exercise-induced angina
- Other heart-related parameters

The target variable indicates whether a patient is at risk of heart attack (1) or not (0).

The dataset was examined for missing values, data types, and class distribution before preprocessing.

3. Methodology

3.1 Data Preprocessing

The following preprocessing steps were performed:

- Identified target and feature variables
- Handled missing values (if any)
- Encoded categorical variables
- Split dataset into training and testing sets
- Applied feature scaling for Logistic Regression

3.2 Model Development

Two supervised learning models were implemented:

1. Logistic Regression
2. Random Forest Classifier

Logistic Regression was used as a baseline model due to its simplicity and interpretability. Random Forest was selected as a more advanced ensemble model capable of handling non-linear relationships.

4. Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC Curve

Results:

- Logistic Regression Accuracy: ~85%

- Random Forest Accuracy: ~87.5%

Random Forest outperformed Logistic Regression in overall performance and provided better balance between precision and recall.

The ROC curve demonstrated strong classification capability, and the confusion matrix helped identify false positives and false negatives.

5. Feature Importance & Explainability

Feature importance analysis from the Random Forest model revealed that variables such as:

- Age
- Cholesterol
- Maximum heart rate
- Chest pain type

had the most influence on prediction outcomes.

This enhances model interpretability and aligns with known medical risk factors.

6. Risk Categorization

To make the system more practical, predicted probabilities were categorized into:

- Low Risk
- Medium Risk
- High Risk

This approach allows easier interpretation in real-world healthcare scenarios and supports clinical decision-making.

7. Error Analysis

Error analysis was conducted by examining false positives and false negatives.

Special attention was given to false negatives (high-risk patients predicted as low risk), as they pose significant medical risks. Understanding these misclassifications helps identify model limitations and potential areas for improvement.

8. Conclusion

The Random Forest model achieved the highest performance with approximately 87.5% accuracy. Feature importance analysis provided valuable insight into key health indicators contributing to heart attack risk.

While the model demonstrates promising results, it should be considered as a decision-support tool rather than a replacement for medical professionals.

9. Future Work

Future improvements may include:

- Using larger and more diverse datasets
- Incorporating ECG time-series data
- Hyperparameter tuning
- Building a web-based prediction interface